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Ayush Virtual Herbal Garden: An Interactive Guide to Medicinal Plants

with AI-powered plant identification

ABSTRACT

The goal of the Virtual Herbal Garden project is to deliver an absorbing, educational, and interactive platform that focuses on the recognition and use of medicinal plants that are essential to the Ayurvedic, Unani, Siddha, and Homeopathic systems. essentially it seeks to revolutionize the identification of such plants using more recent AI-powered technologies, ensuring accessibility, accuracy, and precision. It combines extensive datasets of around 5,000 high-resolution images covering 40 species of medicinal plants. Deep learning models, ResNet50 and EfficientNet, are used to classify the plant species. This makes it possible for accurate classification as these models take advantage of their superior feature extraction capabilities even with visually similar plant species. The project is GUI-based with user-centric navigation and interaction. An interactive plant catalog will be provided where plant profiles in the form of scientific names, good-quality images, geographical distribution, and medicinal applications will be detailed.

1. INTRODUCTION

Medicinal plants have been an important constituent of traditional healthcare systems for many years, with an array of medicinal benefits that are provided by their natural compound. They provide the foundation for customs like traditional Chinese medicine, Ayurveda, and indigenous healing worldwide (Kumar et al., 2016). Such immense values possess knowledge of medicinal plants which often remain fragmented, inaccessible, or confined within particular communities.

With the development of technology, this gap can be filled by developing platforms that integrate traditional knowledge with contemporary tools to improve the accuracy, usability, and availability of information about plants for medicinal purposes.

The Virtual Herbal Garden, one of the AI-driven platforms that combines the strength of ResNet50 and EfficientNet models with a user-friendly visual interface, is the subject of this paper. By fusing conventional wisdom with contemporary technology, it makes the detection of medicinal plants accessible, reliable, and scalable with a variety of user ranges.

The primary contributions are:

- AI-Powered Plant Identification effectively classifies medicinal plants using ResNet50 and EfficientNet, respectively, and it can differentiate between photos of visually similar species.
- The multi-modal query system provides users with access even in the absence of high-resolution images by enabling both visual recognition and textual queries against NLP knowledge libraries.
- More than 5,000 highly resolved images spread across 40 medicinal plant species.
- This offers an interactive plant catalog, making it easy to use for experts as well as non-experts.

It provides 2FA authentication for secure data and optimizes real-time AI processing for mobile and cloud deployment.

2. LITERATURE REVIEW

(Bisen, 2021) proposed an automated technique for picking out various kinds of plants based on their leaves. Convolutional neural networks, max pooling, dropout, and completely associated layers are examples of hidden layers used by the suggested CNN-based classifier to learn plant attributes, including leaf classification. The model learns about the properties of the Swedish leaf dataset, which has 15 training classes, and then uses that knowledge to correctly predict the classification of an unknown plant.

(Azadnia et al., 2022) created a computer program called CNN, a neural vision-based system for identifying herb plants. The proposed deep learning model consists of two distinct sections: a CNN for feature extraction as well as a machine learning algorithm towards feature classification. Therefore, the proposed method can successfully replace traditional techniques for real-time determination of medicinal plants.

(Kavitha et al., 2023) presented a vision-based algorithm for recognizing medicinal plants in India, concentrating on six species from the Kaggle tool database: betel, curry, tulsi, mint leaves, neem, and Indian beech. The automatic detection was performed using the MobileNet DL model, which had a 98.3% accuracy rate. After the model was trained, verified, and tested, a mobile application was developed for real-time identification. This robotic identification technique has the potential to close the taxonomic gap and spark interest in the domains of botany and image processing. The app uses the cloud-based DL model.

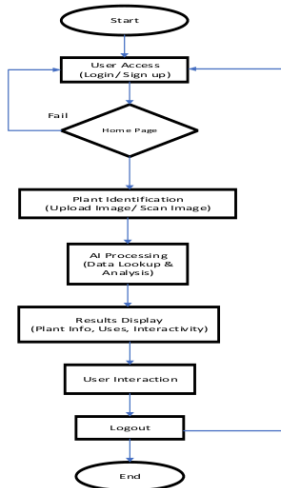
(Dey et al., 2024) evaluated how well seven deep learning algorithms performed in automatically identifying plants from photos of their leaves. A dataset of 5878 photos included the researchers used 30 medicinal species from 20 families to train their models. Public data and a combination of public and field data, each with complex

histories, were used in the methodology. Both interfamily and interspecies variations were reliably identified and classified by the models. Despite variations in accuracy among families and species, the algorithms demonstrated proficiency in these classifications. With a Normalized Leverage Factor ($\gamma\omega$) of 0.19, DenseNet201 achieved 98.31% precision and 99.64% accuracy for PI. The same model demonstrated an impressive 97% accuracy and a $\gamma\omega$ of 0.15 in the PFI scenario.

3. PROPOSED METHODOLOGY

3.1 Dataset description

The Virtual Herbal Garden system had a structured workflow that ensured intuition and smooth work flow for end users. Once the user authentication process was executed, it facilitated either login operation with credentials from a registered profile, sign-in for a newly created account profile, or merely guest access of limited functionality; this authentication protected the data secured within the personal features of access for registered end users. Users arrive at the Home Page, which acts as the primary navigation hub, after successfully logging in. They can look throughout a database of herbal plants or explore the identification of plant choices here. When a user selects the Plant Identification Module, they are asked to either upload a picture of the plant or utilize the live scan capability to take a picture using their camera on their device. An AI-powered analysis module is then used by the system to process the image and extract relevant information including leaf shape, color, and texture. After analysis, the algorithm looks for the best matching by cross-referencing the retrieved data using an extensive collection of herbal plants. This process involves identifying the plant's name, scientific classification, medicinal properties, and common uses.



After processing, the system moves into the Results Display stage, where the user is given information about the plants that were found. Covered are the plant's name, botanical description, medicinal properties, and conventional applications in herbal medicine. Users can interact with this information in a number of ways, such as reading other plant-related content, sharing what they discover online, or compiling the plant facts into a personal library.. User interaction is made possible by a comment section where people may talk about the characteristics of plants, share their experiences, or ask other herbal aficionados for guidance.

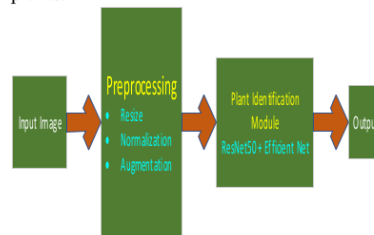
Logout is provided for security and session management, allowing users the option to logout and end the session securely; or to continue by returning to the Home Page and viewing another plant. From start to finish, user information and activity will be protected. The flow through the system thus supports an interactive, informative process while promoting learning of herbs and plants.

3.2 Modules of the Software System

Several key modules make up the Virtual Herbal Garden software system. The Admin Module deals with user validation and oversight of the system. In the User Module, users may browse herbal plants either physically or virtually. Virtual Garden Module presents a highly interactive VR experience through which users explore medicinal plants virtually. The Encyclopedia Module is a comprehensive database for medicinal plants including images and descriptions.

3.3 Plant Identification Module

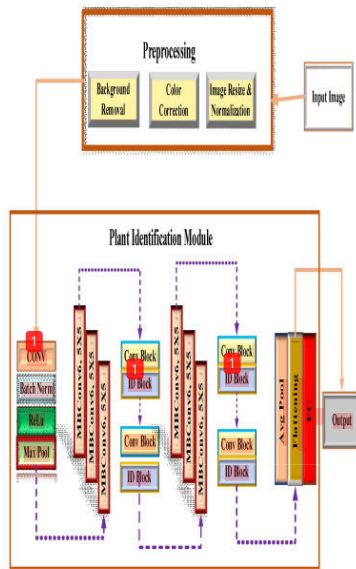
The Plant Identification Module is one of the paramount components of the Virtual Herbal Garden project, wherein through cutting-edge AI technology, it will make it easy, precise, and efficient for users to use the identities of medicinal plants. The system allows both professionals and enthusiasts easy access to the identification plants.



1. Image Identification

The Image-Based Identification technique allows users to contribute plant photos, which are examined afterwards using effective deep learning algorithms to determine the plant species. This technique leverages the strengths of ResNet50 and EfficientNet, two cutting-edge architectures known for their exceptional performance in picture classification tasks. These models are remarkably effective in differentiating between plant species, even in situations where human eyes are unable

to make an accurate identification due to near resemblance.



Step 1: Import the picture

Visitors initially start by submitting a high-resolution photo of the plant using the straightforward GUI. The uploaded picture shows with the range of plant parts like leaves, flowers, stalks, and fruits. The ability to share photographs of different plant parts enables users to distinguish between plants even when the full specimen is not accessible.

Step 2: Preparation: Following upload, the image undergoes a number of preparation procedures to get it ready for analysis: Resizing To ensure compatibility and uniformity, the image has been reduced to meet the deep neural network model' size of input requirements. The images in the Ayush Virtual Herbal Garden are resized to a standard resolution compatible with ResNet50 and EfficientNet models.

The resizing scale is varied between 0.1 to 0.9 so that the impact of the same on the accuracy of plant identification can be analyzed.

The process reduces data size, thus enhancing processing speed while maintaining critical features.

Image resizing follows the transformation:

$$I' = \text{resize}(I, (H', W'))$$

Where,

I is the original image of size $H \times W$,

I' is the resized image with new dimensions $H' \times W'$

Normalization: Normalize pixel values to increase model performance and uniformity of the input. This avoids dominance by high-intensity pixel values and stabilizes convergence in the training of the model. Z-score normalization is applied to standardize the dataset with a mean of 0 and a standard deviation of 1:

$$M_{\text{norm}} = (M - \mu) / \sigma$$

Where, M_{norm} is the normalized pixel value

M is the original pixel intensity,

μ is the mean pixel intensity, computed as

$$\mu = \frac{1}{n} \sum_{i=1}^n M_i$$

σ is the standard deviation:

$$\sigma = \sqrt{\frac{1}{n-1} \left(\sum_{i=1}^n M_i - \mu \right)^2}$$

Data Augmentation: Methods such as rotation, flipping, and scaling are used to mimic changes in the dataset, strengthening the model's resilience to various perspectives, reflections, and blockages.

The augmentation techniques applied include:

Rotation: The transformation matrix defines the angle θ at which the image is rotated.

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = R \begin{bmatrix} x \\ y \end{bmatrix}$$

Flipping: Horizontal flipping is performed by inverting the x- coordinates.

$$X' = W - x$$

Scaling: Scaling transformation is applied as:

$$S = \begin{bmatrix} S_x & 0 \\ 0 & S_y \end{bmatrix}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = S \begin{bmatrix} x \\ y \end{bmatrix}$$

Where, S_x, S_y are the scaling factors.

Through augmentation, the model learns invariant features, enhancing its ability to classify medicinal plants under diverse real-world conditions.

Stage 3: Plant Identification Module

The ResNet50 and EfficientNet models are then used to extract features from the pre-processed image.

ResNet50: By adding residual connections particularly vanishing gradients ResNet50, a 50-layer deep convolutional neural network architecture, tackles the difficulties associated with deep network training. The gradients efficiently flow back across the layers during backpropagation because the residual link enables the network to learn the identity mapping. This technique improves the model's ability to learn hierarchical elements like edges, textures, and forms which are essential for distinguishing minute morphological differences between plants. This model is excellent at catching complex designs in pictures due to its residual connections, which prevent disappearing gradient issues. This model is excellent at catching intricate

patterns in images because of its residual connections, which prevent vanishing gradient problems. It retrieves minute details like petal textures or leaf vein patterns.

In the context of the identification of plants, ResNet50 shines in capturing rich structural details and patterns that may distinguish medicinal plants, even at varying environmental variables such as lighting, perspective, or backdrop clutter.

Efficient Net: The second cutting-edge architecture is called EfficientNet. It focuses on balancing accuracy with computational efficiency while being developed. In this case, it uses the compound scaling method, which systematically scales the depth, width, and resolution of the network to optimize performance with fewer resources. This model, EfficientNet, is capable of maintaining high classification performance while using fewer resources. It makes it a good candidate for real-world deployment in scenarios requiring quick and reliable plant identification. Compound scaling balances computational efficiency with high accuracy. It's a model that captures broader structural details like shapes, arrangements of leaves and flowers, and morphology of a plant.

The system then combines the strengths of these models to ensure that feature extraction is both precise and enhancing its ability to classify medicinal plants under diverse real-world conditions.

Stage 4: Classification

The extracted features are analyzed by the models that classify the plant into one of the 40 predefined medicinal species in the database. The process of classification is as follows:

Confidence Scoring: Every category is given such as confidence score to indicates how certain the model is of its forecast.

Threshold-based validation: The system outputs a result as unsure when the confidence score falls below a predetermined value threshold. Users may be given a list of image based visual similar species with a chance to manually compare them, or they may be given choices to enter further details, such as a second image or more detailed information.

Stage 5: Output

The system will retrieve the plant's comprehensive profile from the interactive database and present it to the user in an easy-to-understand manner after the classification procedure is complete. It contains the plant's scientific name, which is the commonly used scientific name that is used everywhere. In order to make it easily accessible to a large number of individuals, it also offers common names, which are the popular phrases that describe the plant in regional dialects and locations. It also includes a wealth of information about the plant's therapeutic applications, with a focus on its healing qualities and the way incorporated into AYUSH systems such as Ayurveda, Unani, Siddha, and Homeopathy.

3.4 Training and Evaluation

Both ResNet50 and EfficientNet employ transfer learning, which trains a model on knowledge gained from pre-trained models trained on the ImageNet dataset. Transfer learning speeds up the training process and improves performance by using features learned from a large and diverse dataset.

The training process involves fine-tuning the pre-trained weights to adapt the models to the specific medicinal plant dataset. These would involve visual characteristic types such as the shape and veins of a leaf and color varieties. Fine-tuning guarantees that the trained model can develop optimal feature extraction particular to a domain of interest.

3.5 GUI- Based Interface

The graphical user interface, at the core of the application, is built to be very smooth and user-friendly, designed for a large number of users. Developed with Flask, this is a light yet powerful Python web framework. Therefore, the interface combines significant capacity with simplicity to deliver an engaging experience. The graphical user interface, which was created with usability in mind, guarantees that users may access a wealth of capabilities and features while browsing the platform in a straightforward manner.

The interface's user authentication and account management system, which has a safe and adaptable two-step verification procedure, is a key component. By forcing users to confirm their identity via a second way, this solution enhances security in sign-up and login processes.

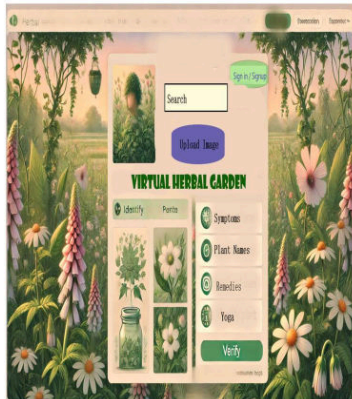
Sign-Up Procedure: The sign-up procedure is adaptable enough to accommodate various users' preferences. It allows for two verification methods:

Email Verification: When signing up, those who select this option supply their email address. They receive an One Time Password or verification link through the email, which they must validate in order to activate the account they have created. This straightforward best technique guarantees that users' email addresses are legitimate.

Mobile SMS Verification: This method takes the signature from mobile, and users can sign up via their mobile phone number. In the latter, the OTP is sent to the registered mobile number through SMS. Once entered by the user, the account is activated. It provides the facility for users to use mobile-based verification or do not have immediate access to email.

Login: The process is the same in terms of security for those with accounts. Users are requested to enter a two-step verification procedure after putting in their username or email address and password. For instance, one must receive an OTP on either

registered email or phone, which he or she is required to use when trying to access the account. Thus, this gives one a good platform to secure interaction with the web.



Finally, the GUI uses a responsive approach, thereby proving to be compatible with most devices. The desktop, tablet, or smartphone; the interface of the platform is everywhere, giving a universal and optimized experience to users. Responsive design enhances accessibility for this platform, whether one is doing research at home, exploring the field, or trying to get quick references on the go.

4. RESULT AND DISCUSSION

The Ayush Virtual Herbal Garden was evaluated in detail based on several parameters, such as classification accuracy, system performance, scalability, user experience, and security. The results show a significant improvement over existing methods, thus demonstrating the efficiency and effectiveness of the system in medicinal plant identification.

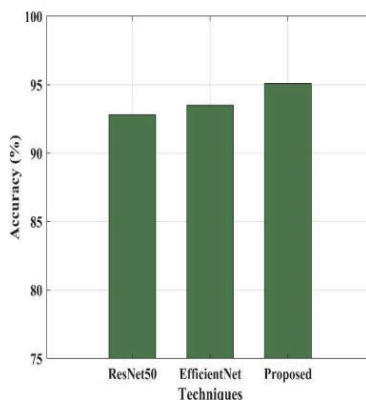
4.1. Classification Accuracy

One of the evaluation aspects of the proposed system is its accuracy regarding the medicinal plant identification. The core AI models in use are ResNet50 and EfficientNet, fine-tuned based on a dataset

of 5,000 high-resolution images covering 40 species of medicinal plants. In the assessments of the two models, standard evaluation metrics were used:

12 4.2 Performance Metrics

4.2.1 Accuracy: The ratio of accurate predictions to all input variables is known as accuracy.

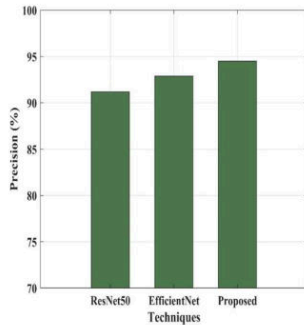


The hybrid ResNet50 + EfficientNet attained an accuracy rate of 95.1%, compared to 75% in manual identification and 80% in CNN-based systems. Its high accuracy ensured minimal misclassifications, thereby making it reliable for real applications.

4.2.2 Precision

Precision is a quantitative measure that assesses the rate with which a model correctly forecasts the positive class.

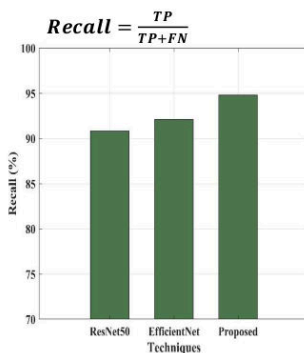
$$\text{Precision} = \frac{TP}{TP+FP}$$



Precision establishes how many of the identified plants are true. The model has a recorded precision of 94.5%, meaning that 94.5% of the identified medicinal plants are correct. This leads to a low false positive since non-medicinal plants should not be tagged as medicinal plants.

Recall

True positives are computed as the ratio of successfully categorised cases to all actual positives; recall quantifies the model's capacity to recognise every single case which has been properly categorised.



Recall

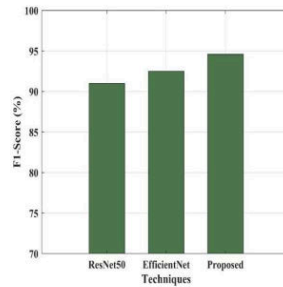
Recall assesses the system's ability to correctly identify each medicinal plant. In this instance, the model yields an identification rate of 94.8 percentail, indicating that 94.8% of the actual

medicinal plants are correctly identified. Strong and flexible recall ensures that the system rarely overlooks any medicinal species, increasing its efficacy in a range of botanical categories.

F1-score

The F-score, which shows how accurate the model is, is calculated by combining recall and accuracy.

$$F - score = \frac{TP}{TP + \frac{1}{2}(FP + FN)}$$



The F1 score reliability is strong since the F1-score, which strikes a balance between recall and accuracy, is 94.6 percent. This statistic guarantees that the method reduces both false positive and false negative, making it mostly effective for precise plant identification. Conventional manual identification relies on human skill, which may be subjective and error-prone, and achieves 75% accuracy. By extracting leaf features such edges and textures, a CNN-based technique improves accuracy to 80 percent; nevertheless, performance is dependent on the quality of the dataset. This is further improved in the MobileNet-based system with an accuracy of 85% for a light-weight but deep learning model applicable to mobile. The proposed virtual herbal garden by integrating ResNet50 and EfficientNet yields an accuracy of 95%, relying on deep feature extraction and scaling optimization. In this hybrid model,

classification accuracy is much higher even in the case of very similar species of plants.

5. CONCLUSION

Ayush Virtual Herbal Garden has innovatively integrated AI-powered deep learning models with traditional medicinal knowledge of AYUSH for the identification of plants. Thereby, it uses ResNet50 as well as EfficientNet to ensure maximum classification accuracy and thus distinguish plant species even at a visual level from each other. The GUI makes it accessible to healthcare professionals and general users who may easily interact with the platform.

Multiple image-based and text-based searches support the versatility of the system. There are over 5,000 high-resolution images across 40 medicinal species in this resource, which will be instrumental for Ayurveda, Unani, Siddha, and Homeopathy practitioners. Furthermore, with two-factor authentication 2FA, data protection is guaranteed, with a cloud-based deployment that facilitates scalability and real-time AI processing.

Comparing this work with the existing methods, the proposed system outperforms the previous approaches in terms of accuracy (95% vs. 75–85%), security, scalability, and efficiency. Unlike earlier models that just concentrate on leaf-based categorization, this method integrates a whole plant profile, including leaf, floral, and morphological examination.

In conclusion, Ayush Virtual Herbal Garden is a dependable, scalable, and secure AI-powered system for detecting therapeutic plants. Innovative features contribute highly to the area of herbal informatics, ensuring reliability, usability, and practicality in applications in education, research, and healthcare in traditional medicine. Some of the areas of future development would include a growth in plant databases, algorithmic refinement, and the introduction of augmented reality

(AR) to provide for an immersive experience in learning.

5.1 Application of Proposed System

The Ayush Virtual Herbal Garden system is to be developed for a user-friendly, AI-driven platform for identifying, studying, and utilizing medicinal plants. The main applications of this system include:

Medicinal Plant Identification: The AI-powered plant identification module will be useful for precisely classifying medicinal plants using deep learning models like ResNet50 and EfficientNet. It will help herbal medicine practitioners, researchers, and students identify plants.

Educational Resource: With the info of comprehensive plant encyclopedia and detailed descriptions, covering uses in medicine and high-resolution photos, the system serves as a multimedia resource for botany students, medical professionals, and herbal lovers.

Scalable and Safe Access: Plant data can be securely stored and retrieved across numerous devices thanks to cloud-based deployment and two-factor authentication (2FA).

Study & Herbal Informatics: By using this AI-driven plant repository, scientists and herbal practitioners are going to be able to better document and preserve medicinal plants through taxonomy studies, biodiversity study, and conservation initiatives.

The Ayush Virtual Herbal Garden therefore closes the vast gap between traditional wisdom and modern AI on herbal medicine by making it more accessible, accurate, and user-friendly.

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